Does quantum mutual information predict when a quantum reservoir beats a tuned classical one?

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Quantum reservoir computing (QRC) uses the dynamics of a driven quantum system to process time series, training only a linear readout. Which property of the quantum state makes a reservoir useful is still debated: entanglement, coherence and total correlations have each been linked to performance, but almost always to absolute performance rather than to the margin over a carefully tuned classical baseline. We ask whether the quantum mutual information of the input-driven reservoir state, computed without access to task targets, predicts the NARMA-10 performance gap between small transverse-field Ising reservoirs and capacity-matched, tuned echo state networks, beyond what entanglement negativity and ℓ_1-norm coherence already explain. We compare this relationship under pure dephasing and amplitude damping, include finite-shot noise throughout, and test whether a fitted diagnostic transfers across decoherence channels, system sizes and tasks. [TODO: Summarise the main results and their implications once Sec. V is complete.]

I. INTRODUCTION

Many prediction problems in science and engineering involve time series. Solving them requires a model that remembers past inputs and transforms them nonlinearly. Reservoir computing (RC) meets this need by driving a fixed, high-dimensional dynamical system with the input and training only a linear readout of its state [1–3]. Because the reservoir is never trained, it need not be a simulated network. Any physical system with fading memory and a suitable nonlinear response can serve, which has made RC a leading framework for computing with unconventional hardware [4, 5].

Quantum systems are an attractive substrate. An N-qubit reservoir evolves in a 2^N-dimensional state space, and its dynamics can generate correlations with no classical counterpart [6, 7]. Quantum reservoir computing (QRC) needs no gradient-based training of the quantum dynamics, and it requires some dissipation to obtain fading memory [6, 8]. Both features suit today's noisy intermediate-scale devices [9, 10]. Since the proposal of Fujii and Nakajima [6], QRC has been studied in spin networks, coupled oscillators, single qudits and superconducting processors. Its performance has been linked to the dynamical regime, the readout observables and the effective dimension of the system [11–16].

It remains unclear which property of a quantum reservoir is responsible for good performance, and whether that property is quantum at all. Entanglement can support short-term memory [17], but its benefit depends on the input frequency once dissipation is present [18], and memory can peak well below maximal entanglement [19]. In a transverse-field Ising reservoir, information processing capacity (IPC) rose monotonically with ℓ_1-norm coherence, while negativity could fall by more than two-thirds without a significant loss of capacity [20]. In the same system, purely classical correlations sustained capacity in the localised regime. Recent preprints propose further candidates, including magic, non-Markovianity, Krylov observability and a memory–nonlinearity balance [21–24]. Because these diagnostics are computed from the reservoir state alone, one that reliably predicted performance could guide reservoir design before any task is attempted. Which measure, if any, can play that role is not settled.

Two factors complicate the search for such a diagnostic. The first is the type of decoherence. Non-unital amplitude damping can enlarge processing capacity and improve performance, whereas dephasing and depolarising noise degrade it [14, 25, 26]. Whether the link between a diagnostic and performance also depends on the channel is a separate question. Bit-flip and phase-flip noise produced a single capacity–coherence curve in Ref. [20], but amplitude damping has not, to our knowledge, been compared in this way. The second factor is measurement. Conclusions drawn from exact expectation values can change once finite sampling is included, and omitting shot noise can reverse the apparent effect of decoherence on performance [20, 27, 28].

The choice of benchmark raises a further problem. Claims of quantum advantage have often been made against classical reservoirs tuned less carefully than the quantum model [13, 16, 29]. When echo state networks (ESNs) were matched in feature dimension and tuning budget, the advantage of two QRC mechanisms disappeared in noiseless simulations of up to 11 qubits [30]. A recent survey concludes that no broad advantage over well-matched classical reservoirs has yet been established [31]. For the study of quantumness diagnostics, this means that relating quantum properties to absolute measures, such as memory capacity or IPC, is not sufficient. The quantity of practical interest is the gap between a quantum reservoir and a fair classical baseline, and the question is whether a property of the quantum state can indicate in advance when that gap is positive.

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We address this question for small transverse-field Ising reservoirs on the NARMA-10 benchmark [32]. We ask whether the quantum mutual information (QMI) of the input-driven reservoir state, computed without access to task targets, explains variation in the performance gap to a matched, tuned ESN beyond what negativity and ℓ_1-norm coherence already explain. We also ask whether this relationship differs between pure dephasing and amplitude damping. QMI is a natural candidate. It is cheap to compute from the state and captures total correlations, both classical and quantum [33]. In Ref. [20] it closely tracked coherence in most regimes, while classical correlations dominated in the localised regime. QMI may therefore carry information that coherence and entanglement miss in some regimes, and be redundant with them in others. Our design is built to tell these outcomes apart.

The study makes four contributions:

1. It tests, using nested cross-validated models and equivalence tests, whether QMI adds predictive information about the quantum–classical performance gap beyond negativity and coherence (H1a).

2. It compares the diagnostic–performance relationship under amplitude damping and pure dephasing, which to our knowledge has not been done before (H1b).

3. It tests whether a diagnostic model fitted in one setting transfers to another channel, a larger system ($N = 8$) and a second task (H1c).

4. It includes finite-shot noise in every primary analysis and uses a classical baseline built to current fair-comparison standards [30, 31]. As a validation step, it first reproduces the IPC–coherence and IPC–negativity results of Ref. [20].

Section II reviews related work. Section III states the hypotheses. Section IV describes the reservoir model, the decoherence channels, the diagnostics, the classical baseline and the statistical analysis. Section V presents the results for each hypothesis, and Sec. VI discusses their implications and limitations.

II. BACKGROUND AND RELATED WORK

A. Reservoir computing and fading memory

Reservoir computing keeps a recurrent dynamical system fixed and trains only a readout, usually by linear regression [1, 31, 34]. The approach needs fading memory: the reservoir state must depend on recent inputs and forget its initial condition, a requirement known as the echo state property [1]. Information processing capacity makes this quantitative [35]. It measures how well the reservoir reproduces orthogonal polynomials of delayed inputs, and its total is bounded by the number of linearly independent output features.

NARMA tasks are a standard benchmark because they need both memory and nonlinearity [32]. NARMA-10, in particular, requires the reservoir to combine inputs from ten steps back. Recent work on a damping-controlled quantum reservoir found that NARMA performance peaked where memory capacity was highest, and proposed memory capacity as a practical indicator of performance [26].

B. Quantum reservoir computing

QRC replaces the classical network with a quantum system. The standard model, introduced by Fujii and Nakajima [6], is a transverse-field Ising network: an input qubit is reset and rewritten at each step, the network evolves, and local observables are read out with time multiplexing. Because unitary evolution is reversible, fading memory requires dissipation; the reset-and-write step supplies it [20]. Dissipation contracts differences between initial states, but its rate matters. Contraction that is too fast erases recent inputs, and contraction that is too slow leaves a dependence on initialisation [31]. Engineered local losses can themselves be a resource, and dissipative spin models form a universal class for fading-memory maps [8].

The nominal state space grows as 2^N, but this overstates what the readout can use: only features exposed by the encoding, dynamics and measurement contribute [31]. Within that limit, the dynamical regime matters. Ergodic dynamics favour tasks needing nonlinear memory, while many-body localisation restricts information flow [11, 20]. Readout design also matters: two-body correlators raise total IPC over single-spin observables [12]. A single d-level system improves as its dimension grows and has outperformed a physically analogous classical reservoir [13]. Reviews [7, 31] list further proposed explanations for QRC performance, from dissipation to chaos and measurement back-action.

C. Quantumness diagnostics and performance

Entanglement. Götting *et al.* linked entanglement and the occupied phase-space dimension to linear short-term memory. They concluded that high entanglement is a prerequisite for the richer dynamics that make use of the exponential state space [17]. Kora *et al.* measured entanglement by logarithmic negativity and found that its benefit depends on the input [18]. Without dissipation the entanglement advantage was always present. With dissipation, fast-varying signals gained more from entanglement, and slow signals were better served by a non-entangled reservoir. The benefit also saturated at high entanglement [18]. In a spin network with two in-

put qubits, memory capacity peaked at moderate coupling, well before average entanglement reached its maximum [19].

Coherence. A synthetic reservoir with tunable coherence improved on short-term-memory and parity-check tasks as coherence increased [29]. Palacios *et al.* found a monotonic relationship between IPC and ℓ_1-norm coherence in the ergodic regime of the transverse-field Ising reservoir [20]. Negativity could fall by more than two-thirds of its noiseless value before capacity dropped significantly. The authors read this as superposition mattering more than entanglement for that reservoir.

Total correlations. Palacios *et al.* also computed the total mutual information, the quantum hookup and purely classical correlations [20]. The hookup equals total correlations plus local coherence [36]; local coherence was zero in their system, so the hookup and mutual information coincided. Empirically, both closely tracked ℓ_1 coherence whenever classical correlations were small. Classical correlations carried most of the remaining capacity in the localised regime under strong phase-flip noise. In coupled Kerr oscillators, time-averaged mutual information fell as dissipation rose, while long-term memory capacity improved through more redundant encoding [37]. Correlations and performance can therefore move in opposite directions, depending on the regime.

Other proposed resources. Recent preprints propose further explanations. Dynamics-generated magic is argued to be necessary for computation in qubit-resetting schemes, with the echo state property enforced separately by the Liouvillian spectral gap in Hamiltonian-encoding schemes [22]. Markovian architectures are argued to forget exponentially, and non-Markovian reservoirs are shown to outperform them on tasks mixing short- and long-term correlations [23]. Krylov observability explained Lorenz-63 performance better than IPC when the reservoir was undersampled [24]. Memory restriction, weak measurement, the edge of chaos and dissipation have been unified as tunings of a single memory–nonlinearity balance measured by IPC [21]. Against these, QMI is cheap to compute from the state and is the natural total-correlation quantity, which is why this study focuses on it.

D. Decoherence channels

The evidence favours amplitude damping over unital noise for performance, but mostly on tasks other than NARMA-10. Domingo *et al.* showed that amplitude damping improved a quantum-chemistry regression task (the first excited-state energy of LiH) at error rates of about 5×10^{-4} or below and shallow depth, while depolarising and phase damping always degraded it [25]. Their explanation is that a non-unital channel adds new Pauli components, whereas unital channels only shrink existing ones. They name time-series forecasting as future work, so this result is not a temporal one.

For temporal tasks, Kubota *et al.* found that amplitude damping produced time-invariant processing capacity that phase-flip, bit-flip, phase-damping and depolarising noise did not, and that capacity rose with noise level; they confirmed this on IBM hardware [14]. Ricci *et al.* reported that controllable amplitude damping supports NARMA and Mackey–Glass tasks, while dephasing and depolarising noise progressively degrade memory capacity [26]. The native noise of superconducting processors has also served as the dissipation source for temporal benchmarks [15]. For dephasing, Fujii and Nakajima suggested a constructive role in the reset-and-write map [6, 20], and Götting *et al.* discuss its effect on performance [17].

Two cautions follow. First, Palacios *et al.* report that some non-monotonic noise effects found earlier do not survive realistic measurement noise [20]. Second, performance may differ by channel while the diagnostic–performance relationship does not: bit-flip and phase-flip data collapsed onto one capacity–coherence curve [20]. These are separate questions, and H1b addresses the second.

E. Finite sampling

Real reservoirs are read out from a finite number of measurements, and this changes conclusions. Statistical noise can lower total capacity [28, 38]; frameworks for finitely sampled quantum reservoirs are developed in Refs. [39, 40]. Palacios *et al.* modelled roughly 10^6 shots as Gaussian noise with standard deviation 10^{-3}, and showed that omitting it can reverse the apparent effect of decoherence strength on performance [20]. Kora and Simon found that statistical noise harms performance overall but that entangled, coherent reservoirs are more robust to it than unentangled, incoherent ones, so the benefit of quantumness depends on the number of measurements [27]. This study therefore includes shot noise in every primary analysis.

F. Fair comparison with classical reservoirs

Advantage claims depend on the strength of the classical baseline. A July 2026 survey concludes that current results do not establish a broad quantum advantage over well-matched classical reservoirs, and calls for matched baselines and complete resource accounting [31]. A companion study tuned classical baselines as carefully as the quantum models in noiseless simulations of up to 11 qubits [30]. Both QRC mechanisms tested, two-point correlator readout and feedback-driven QRC (cf. Ref. [41]), lost their advantage. On a hidden Markov-switching autoregressive prediction task, the tuned ESN reached a normalised RMSE of 0.720 ± 0.060 against 0.787 ± 0.059 for feedback QRC, over 10×10 seeds. The author limits the claim to these simulable sizes. Earlier advantage

claims against ESNs [16, 29] and against a physically analogous classical system [13] should be read against this standard.

G. Research gap

The literature relates quantumness to absolute performance, not to the gap against a fair classical baseline. Palacios *et al.* [20] already compare mutual information, coherence and negativity against IPC, so the novelty here is narrower. Table I summarises what remains open and how each hypothesis addresses it.

III. RESEARCH QUESTION AND HYPOTHESES

Research question. In small spin-based quantum reservoirs under local decoherence, does the QMI of the input-driven reservoir state, computed without task targets, explain variation in the NARMA-10 performance gap to a matched, tuned ESN beyond what entanglement negativity and ℓ_1-norm coherence explain? Is that relationship the same under pure dephasing and amplitude damping?

Definitions. The performance gap Δ (Sec. IV F) is the ESN's NRMSE minus the quantum reservoir's NRMSE, so $\Delta > 0$ means the quantum reservoir is better. Diagnostics are time-averaged over the input-driven state after washout and never use the targets. The primary QMI is the multipartite total mutual information $\mathcal{T}$; pairwise QMI is secondary (Sec. IV G). All performance figures include finite-shot noise. The hypotheses are summarised in Table II.

Rationale. QMI counts classical as well as quantum correlations. In Ref. [20], classical correlations carried most of the remaining capacity in the localised regime under strong phase-flip noise, while negativity behaved unlike the other measures. QMI is therefore most likely to add information in strongly dephased or localised regimes, and least likely in the ergodic regime where it tracks coherence (H1a). Amplitude damping is non-unital and produces processing capacity that unital channels do not [14]; it can improve performance where dephasing and depolarising noise degrade it [25, 26]. Against this, bit-flip and phase-flip data collapse onto one capacity–coherence curve [20], so a channel-independent relationship is plausible (H1b). The value of entanglement depends on input frequency under dissipation [18] and on the number of measurements [27]; a diagnostic that transfers across channels, sizes and tasks would be a stronger result than an in-sample correlation (H1c).

Interpretation of outcomes. H1a together with H1c would establish a transferable pre-task diagnostic. H1a without H1c would give a regime-specific diagnostic, with the regimes mapped. Equivalence on H0a would show that QMI is redundant with coherence and negativity, a clean negative result. A gap that is never positive under the fair baseline would be a fair-comparison result in the sense of Ref. [30].

IV. METHODS

Parameter values marked † are proposals that will be fixed in a pre-registration after the pilot and before the main sweep.

A. Reservoir model

The reservoir is a network of N qubits with all-to-all transverse-field Ising couplings and on-site disorder, following Refs. [6, 11, 20]:

$$H = \sum_{i<j} J_{ij}\,\sigma_i^x\sigma_j^x + \frac{1}{2}\sum_{i=1}^{N}(h + D_i)\,\sigma_i^z, \tag{1}$$

with $J_{ij} \sim \mathcal{U}[-J_s/2,\, J_s/2]$ and $D_i \sim \mathcal{U}[-W,\, W]$. Energies are in units of $J_s = 1$ and times in units of $1/J_s$. The field h and disorder W set the dynamical regime: ergodic dynamics at moderate h and weak W, and localised dynamics at strong W [11]. Each regime point is classified by the mean adjacent-gap ratio $\bar{r}$ of H within one parity sector (about 0.53 for ergodic and 0.39 for localised spectra) [42]. Each (h, W) point is sampled with independent Hamiltonian realisations.

Input injection. At step k a scalar input $s_k \in [0, 1]$ is written into qubit 1 by the reset-and-write map of Refs. [6, 20],

$$\begin{aligned} \rho_k &= \mathcal{E}_{\Delta t}[\,|\psi_{s_k}\rangle\langle\psi_{s_k}| \otimes \mathrm{Tr}_1(\rho_{k-1})\,]\,, \\ |\psi_s\rangle &= \sqrt{1-s}\,|0\rangle + \sqrt{s}\,|1\rangle\,, \end{aligned} \tag{2}$$

where $\mathcal{E}_{\Delta t}$ is evolution for one input period Δt under the dynamics of Sec. IV B. The reset supplies the dissipation needed for fading memory [20]. Because the encoding changes noise robustness and the parity structure of IPC [20], a second encoding (rotation about x) is run as a sensitivity check.

B. Decoherence channels and simulation

Decoherence acts locally on every qubit through the Lindblad master equation

$$\dot{\rho} = -i[H, \rho] + \sum_{i=1}^{N}\left(L_i\rho L_i^\dagger - \tfrac{1}{2}\{L_i^\dagger L_i, \rho\}\right), \tag{3}$$

with the jump operators listed in Table III. With these definitions both channels damp single-qubit coherences at rate $\gamma/2$ ($T_2 = 2/\gamma$), so the main comparison is at equal coherence decay. A secondary analysis matches the channels on equal purity loss per step instead.

TABLE I. Open points in the literature and how this study addresses them.

Open point	Closest existing work	What this study adds	Hypothesis
Target variable	Diagnostics vs short-term memory, parity or IPC [17–20]	Predicts the gap Δ to a tuned, capacity-matched ESN	H1a
Channel	Bit flip and phase flip [20]; oscillator loss [37]	Amplitude damping vs pure dephasing for the diagnostic–performance relation	H1b
Task	NARMA with memory capacity as indicator [26]	NARMA-10 with the three-diagnostic comparison	H1a, H1b
Transfer	Generality for local Markovian noise suggested, tested at $N = 5$ only [20]	Held-out channel, size ($N = 8$) and task	H1c
Measurement noise	Shown to reverse conclusions [20]	Shot noise in every primary analysis	All
Fair baseline	Advantage disappears under matched baselines [30, 31]	A diagnostic of when a positive gap exists	All

TABLE II. Hypotheses, null hypotheses and tests. The margins (0.02 in R^2, 0.70 transfer ratio) are fixed in the pre-registration.

ID	Alternative hypothesis	Null hypothesis	Test
H1a	Adding QMI to a model of Δ containing negativity, coherence and controls raises cross-validated R^2 by at least 0.02	H0a: the gain is below 0.02	Nested models with cross-validation grouped by grid cell; redundancy claimed only if a TOST places the gain inside ± 0.02
H1b	The QMI–Δ slope differs between pure dephasing and amplitude damping	H0b: no channel × QMI interaction	Interaction term, Holm-corrected across QMI variants; TOST on the standardised slope difference to claim channel independence
H1c	A model fitted in one setting keeps at least 70% of its in-setting R^2 on a held-out channel, on $N = 8$, and on a second task	H0c: the transfer ratio is below 0.70 in at least one held-out setting	Transfer ratio = held-out R^2 / in-setting R^2; reject H0c when the lower 95% bootstrap bound is ≥ 0.70

TABLE III. Decoherence channels.

Channel	Jump operator	Unital	Role
Pure dephasing	$L_i = \frac{\sqrt{\gamma}}{2}\,\sigma_i^z$	Yes	Main comparison (H1b)
Amplitude damping	$L_i = \sqrt{\gamma}\,\sigma_i^-$	No	Main comparison (H1b)
Bit or phase flip	Pauli channels [20]	Yes	Reproduction only

States are stored as dense $2^N \times 2^N$ density matrices. Each period Δt is split into n_s second-order (Strang) steps: half a step of the local channel, a unitary step, then half a step of the channel. Each local channel is applied exactly with its Kraus operators. The splitting is checked against the exact vectorised Lindblad propagator for $N \leq 5$ (Sec. IV J).

C. Readout and finite sampling

Within each period, the expectation values $\langle \sigma_i^z \rangle$ and $\langle \sigma_i^z \sigma_j^z \rangle$ are recorded at V equally spaced times (time multiplexing) [6, 12]. This gives $D = V\,[N + N(N-1)/2]$ features plus a bias. Proposed values are $V = 10$ and $\Delta t = 10$.[†]

Finite sampling is part of every primary analysis. Each expectation value $\langle O \rangle \in [-1, 1]$ is replaced by $\langle O \rangle + \varepsilon$ with $\varepsilon \sim \mathcal{N}\big(0,\, (1 - \langle O \rangle^2)/n_{\text{shots}}\big)$, the binomial variance of a ± 1 observable, clipped to $[-1, 1]$. The primary setting is $n_{\text{shots}} = 10^6$, close to the $\sigma = 10^{-3}$ used in Ref. [20]; 10^4 and 10^5 are secondary. Measurement backaction is not modelled: we assume an ensemble or restart protocol, and discuss weak and projective measurement schemes [39] as a limitation.

D. Tasks and capacity

The primary task is NARMA-10 [32]. With inputs u_k drawn independently from $\mathcal{U}[0, 0.5]$,

$$y_{k+1} = 0.3\, y_k + 0.05\, y_k \sum_{i=0}^{9} y_{k-i} + 1.5\, u_{k-9}\, u_k + 0.1. \quad (4)$$

The reservoir receives $s_k = 2u_k \in [0, 1]$. Series that diverge are discarded and redrawn, and the number discarded is reported. Each input realisation has a washout of 500 steps, followed by 2000 training, 500 validation

and 1000 test steps.† Five independent input realisations are used per configuration.

For transfer (H1c), two further tasks use the same pipeline. NARMA-5 and NARMA-20 change the memory demand; NARMA-20 uses the tanh-bounded variant [TODO: state the exact form]. Mackey–Glass one-step prediction ($\tau = 17$) adds a chaotic signal with a different structure.

Information processing capacity is computed following Ref. [35]: products of Legendre polynomials of delayed inputs $u \in [-1, 1]$, up to total degree 3 and delay 20.† Capacities below a threshold set from shuffled-input surrogates are set to zero. IPC uses the same shot-noise model as the tasks. It is used for the reproduction of Ref. [20] and as a secondary covariate that separates memory from nonlinearity [21].

E. Readout and classical baseline

All readouts are ridge regressions on the features plus a bias. The penalty λ is chosen on validation data from 21 log-spaced values between 10^{-10} and 1. Performance is the normalised root-mean-square error on the test set,

$$\mathrm{NRMSE} = \sqrt{\frac{\langle (y_k - \hat{y}_k)^2 \rangle}{\mathrm{Var}(y)}}. \tag{5}$$

The classical baseline is a leaky ESN [1, 3],

$$\boldsymbol{x}_{k+1} = (1 - a)\, \boldsymbol{x}_k + a \tanh(W \boldsymbol{x}_k + \boldsymbol{w}_{\mathrm{in}} s_k + \boldsymbol{b}). \tag{6}$$

Its size equals the quantum readout dimension D, so both models fit the same number of readout weights [30, 31]. W is random and sparse, rescaled to spectral radius ρ. The ESN is tuned with a tree-structured Parzen estimator over leak rate $a \in [0.05, 1]$, $\rho \in [0.1, 1.5]$, input scaling in $[0.01, 2]$ and bias scaling in $[0, 1]$, with 100 trials per (task, D)† judged on validation NRMSE. The selected settings are scored on the test set over 10 weight seeds.

The quantum reservoir receives no comparable tuning: each configuration is fixed and only λ is fitted. This makes the comparison conservative for the quantum side, which strengthens any positive gap. Two reference baselines are also reported: an untuned ESN ($a = 1$, $\rho = 0.9$, unit input scaling) and a linear–quadratic regression on the last 10 inputs, which shows how much of the task needs no reservoir at all.

F. Performance gap

For configuration c with readout dimension D_c, the performance gap is

$$\Delta_c = \overline{\mathrm{NRMSE}}_{\mathrm{ESN}}(D_c) - \overline{\mathrm{NRMSE}}_{\mathrm{QRC}}(c), \tag{7}$$

where bars denote means over the five input realisations, and over the ten weight seeds for the ESN, which sees the same realisations. $\Delta_c > 0$ means the quantum reservoir is better. The relative gap $\Delta_c / \mathrm{NRMSE}_{\mathrm{ESN}}$ is a secondary outcome.

G. State diagnostics

Diagnostics are computed from the reservoir state alone and never see task targets. After washout, the reservoir is driven by a fresh input sequence of 200 steps from the task's input distribution. Each diagnostic is evaluated on the state at the end of every period and averaged over the window. Entropies use base-2 logarithms. The diagnostics are:

$$\bar{\mathcal{N}} = \frac{1}{2^{N-1} - 1} \sum_{A|B} \frac{\|\rho^{T_A}\|_1 - 1}{2}, \tag{8}$$

$$C_{\ell_1} = \sum_{i \neq j} |\rho_{ij}|, \tag{9}$$

$$\mathcal{T} = \sum_{i=1}^{N} S(\rho_i) - S(\rho), \tag{10}$$

$$\bar{I} = \binom{N}{2}^{-1} \sum_{i<j} \left[S(\rho_i) + S(\rho_j) - S(\rho_{ij}) \right], \tag{11}$$

where the sum in Eq. (8) runs over all bipartitions [20, 43], C_{ℓ_1} is evaluated in the computational basis that matches the readout [44], and $\mathcal{T}$ is the multipartite total correlation of Refs. [20, 33]. $\mathcal{T}$ is the primary QMI; $\bar{I}$, the half-chain mutual information and log-negativity [18] are secondary. For comparison across sizes, C_{ℓ_1} is divided by $2^N - 1$ and $\mathcal{T}$ by N. Every model also includes the controls purity $\mathrm{Tr}\,\rho^2$, von Neumann entropy $S(\rho)$, $\log \gamma$, h, W, $\bar{r}$ and N. For pure states, the mutual information across a bipartition equals twice the entanglement entropy [45], so any difference between QMI and entanglement measures must come from mixedness, which the purity control tracks.

H. Experimental design

The unit of analysis is a configuration c: one Hamiltonian realisation at one point of the parameter grid. A grid cell is the set of realisations sharing $(N, \mathrm{channel}, \gamma, h, W)$; cells define the groups for cross-validation. The sweeps are summarised in Table IV. The h and W values are fixed after a pilot that maps $\bar{r}$ across a coarse grid, so that the three levels sit clearly in the ergodic, crossover and localised regimes. The pilot also times one $N = 8$ configuration to confirm the compute budget.

I. Statistical analysis

All predictors are standardised within the fitting set. Cross-validation is 5-fold, grouped by grid cell so that

TABLE IV. Experimental design.

Factor	Main sweep (fitting)	Transfer sweep (H1c)	Reproduction of Ref. [20]
N	4, 6	8	5
Channel	Dephasing, amplitude damping	Dephasing, amplitude damping	Bit flip, phase flip
γ (units of J_s)	8 log-spaced values, 10^{-3} to $1^{\dagger}$	4 of the main values	As in Ref. [20]
h	3 values spanning the regimes†	Same	As in Ref. [20]
W	3 values, weak to strong disorder†	Same	As in Ref. [20]
Realisations per cell	10	5	As in Ref. [20]
Configurations	2880	360	–
Outputs	NARMA-10, IPC, diagnostics	NARMA-10, NARMA-5/20, Mackey–Glass, diagnostics	IPC, coherence, negativity

realisations of one cell never sit on both sides of a split, and repeated 10 times. Uncertainty comes from a cluster bootstrap over cells (2000 resamples), recomputing the full cross-validation in each resample. Each primary hypothesis is tested at $\alpha = 0.05$; they answer separate questions and are not corrected across.

H1a (added value of QMI). Two linear models of Δ are compared,

$$M_0 : \ \Delta \sim \bar{\mathcal{N}} + C_{\ell_1} + \text{controls}, \tag{12}$$

$$M_1 : \ \Delta \sim \bar{\mathcal{N}} + C_{\ell_1} + \mathcal{T} + \text{controls}. \tag{13}$$

The statistic is $\Delta R^2_{\text{cv}} = R^2_{\text{cv}}(M_1) - R^2_{\text{cv}}(M_0)$. H1a is supported if $\Delta R^2_{\text{cv}} \geq 0.02$ and its 95% lower bound exceeds 0. QMI is declared redundant if the 90% interval lies within ± 0.02 (TOST at $\alpha = 0.05$); otherwise the result is inconclusive. The test is run pooled, within each channel, and within the ergodic and localised regimes. Sensitivity analyses swap $\mathcal{T}$ for $\bar{I}$, use log-negativity, and repeat the nested comparison with gradient-boosted trees to allow nonlinear effects. Correlations, variance inflation factors and a commonality analysis show how much of the explained variance QMI shares with coherence.

H1b (channel dependence). On the pooled data we fit

$$\Delta \sim \bar{\mathcal{N}} + C_{\ell_1} + \mathcal{T} + \text{controls} + \text{channel} + \text{channel} \times \mathcal{T}. \tag{14}$$

The channel $\times\, \mathcal{T}$ coefficient is tested with standard errors clustered by cell, with Holm correction across the two QMI variants. Channel independence is claimed only if a TOST places the standardised interaction within ± 0.1.† Interactions of channel with C_{ℓ_1} and $\bar{\mathcal{N}}$ are reported as secondary.

H1c (transfer). M_1 is fitted on a source setting and applied unchanged to a target setting, using size-normalised diagnostics and the source's standardisation. Three transfers are tested: between channels (both directions), from $N = 4, 6$ to $N = 8$, and from NARMA-10 to each second task. The transfer ratio is $R^2_{\text{target}} / R^2_{\text{cv,source}}$. H0c is rejected when the 95% lower bound of the ratio is at least 0.70 in all three transfers. Because Δ can shift in scale between tasks, the Spearman rank correlation between predicted and observed Δ is reported as a secondary measure.

J. Validation

The main sweep starts only after the following checks pass.

1. *Unit tests.* A Bell pair gives $\bar{\mathcal{N}} = 0.5$ and $\mathcal{T} = 2$ bits; $|+\rangle^{\otimes N}$ gives $C_{\ell_1} = 2^N - 1$ and $\mathcal{T} = 0$; product states give zero negativity; each channel's Choi matrix is positive semidefinite and trace preserving; amplitude damping relaxes to $|0\rangle$.

2. *Integrator accuracy.* For $N \leq 5$, the Strang splitting matches the exact Lindblad propagator to trace distance below 10^{-4} after 100 periods; n_s is set by this test.

3. *Echo state property.* For every configuration, two random initial states driven by the same input end the washout within trace distance 10^{-6}. Configurations that fail are reported and excluded.

4. *Reproduction of Ref. [20].* At $N = 5$ under bit-flip and phase-flip noise, the IPC–coherence relation is monotonic, negativity falls substantially before capacity drops, and the two channels collapse onto one curve.

5. *Baseline sanity.* The tuned ESN's NARMA-10 error is compared with published values for networks of similar size and must beat the linear–quadratic baseline.

K. Reproducibility

Hypotheses, margins, the parameter grid and the analysis plan are pre-registered [TODO: OSF link] after the pilot and before the main sweep. All code, configuration files and seeds are public [TODO: GitHub and Zenodo DOI]. Every run records its configuration hash, seeds, code version and package versions. Raw results are stored as Parquet tables, and every figure and table in the paper is produced by a script from those tables.

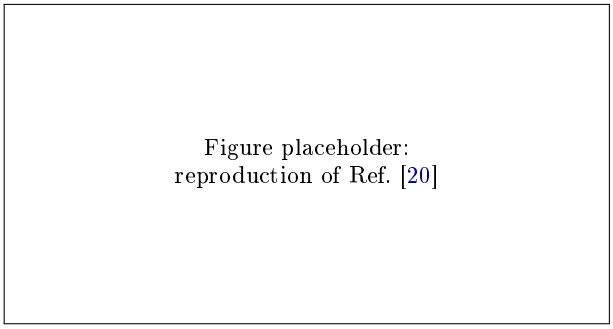


FIG. 1. [TODO: Caption.] IPC versus ℓ_1 coherence, negativity and total mutual information for $N = 5$ under bit-flip and phase-flip noise.

V. RESULTS

[TODO: To be written after the main sweep. Suggested order:]

1. Validation: integrator accuracy, echo-state pass rate, and the reproduction of Ref. [20] (Fig. 1).
2. The gap Δ across channels, noise rates and regimes.
3. H1a: added value of QMI, pooled and by channel and regime.
4. H1b: channel dependence of the QMI–Δ slope.
5. H1c: transfer across channel, size and task.
6. Sensitivity: shot number, encoding, nonlinear models.

VI. DISCUSSION

[TODO: Interpret the results against Sec. II. Limitations to address: small system sizes ($N \leq 8$); the ensemble or restart readout without back-action; the asymmetric tuning budget (conservative for QRC); the choice of channel matching; and the dependence on one Hamiltonian family.]

VII. CONCLUSION

[TODO: Two or three sentences on the answer to the research question.]

DATA AND CODE AVAILABILITY

[TODO: Repository URL, Zenodo DOI and pre-registration link.]

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